Module - 1

① Define Data science and Explain its life cycle.

Data science is an interdisciplinary field that uses scientific methods, statistics, mathematics, programming, machine learning and domain knowledge to collect, Process, analyze and interpret large amount of data to obtain.

Data science Life Cycle:-

The major stage of the data science Life cycle are:

1. Problem Definition:-

The first step is to clearly understand the problem and determine the objectives.

Ex:- A company wants to predict which customers are likely to stop using its service

2. Data collection:-

Relevant data is collected from different source such as.

Ex: Databases, website, surveys, sensors, socialmedia, Applications

3. Data PreProcessing:-

Collected data may contain missing values, duplicate records, error, and inconsistent value. These problems are corrected before analysis

4. Exploratory Data Analysis (EDA):-

The cleaned data is explored to understand patterns, relationships and trends.

Ex: Graphs, Charts and statistical methods are commonly used.

5. Model Building :-

Machine learning or statistical models are developed using the prepared data.

Ex: Regression, decision trees, classification algorithms

6. Model Evaluation :-

The models performance is tested using appropriate evulation measures such as Accuracy, precision, Recall and F1-Score.

7. Deployment :-

After successful evaluation, the model is deployed into a real-world application,

8. Monitoring and maintenance :-

The deployed model is continuously monitored and updated when new data become available.

② Differentiate Descriptive, Diagnostic, predictive and prescriptive Analytics

Data Analytics is the process of Examining data to discover useful information, identify patterns and support decision-making.

Descriptive analytics :-

⇒ Descriptive analytics Explains what happend in the past

⇒ It summarizes historical data using reports, charts, graphs and dashboards

Ex:- A company sold 5,000 mobile phone last month.

Ex: Suppose age are,

20, 21, missing, 23, 24

The missing value can be replaced with the mean or median age.

Data Integration :-

Data Integration combines data collected from multiple source into a single dataset.

Ex:- Combining customer details from a customer database with their purchase from a sale database

Data transformation :-

Data transformation converts data into an appropriate format for analysis.

Normalization, scaling, Encoding, Aggregation.

Ex:- Converting gender values

male → 0

female → 1

Data Reduction :-

Data reduction decreases the amount of data while maintaining important information.

methods :-

⇒ Feature selection

⇒ Sampling

⇒ Dimensionality reduction

⇒ Aggregation

Ex:- Removing unnecessary columns such as serial numbers or IDs.

Data Discretization :-

Data Discretization converts continuous numerical val into categories

Ex:-

⇒ 0 - 18 → child

⇒ 19 - 60 → Adult

⇒ Above 60 → senior citizen

Handling outliers :-

Outliers are value that are significantly different from other values in the dataset

Ex:- If salaries are:

20,000, 22,000, 25,000, 23,000, 10,00000

10,00,000 may be considered an outlier and should be investigated before analysis.